

UNIT-II

LEETCODE EXERCISE

1. Find Users with Valid E-Mails

Problem Summary

You are given a table named Users.

Column Name	Type
user_id	int
Name	Varchar
mail	Varchar (50)

- user_id is the primary key.
- Each row contains the ID and name (which is an e-mail address) of a user.

You need to find users whose e-mail addresses are valid.

Valid E-mail Rules

A valid e-mail:

1. Starts with a letter (a-z or A-Z)
2. Followed by:
 - o letters (a-z, A-Z)
 - o digits (0-9)
 - o underscore (_)
 - o period (.)
 - o hyphen (-)
3. Ends with:

@leetcode.com

Task

Write an SQL query to find all users having valid e-mail addresses.

Return the result table in any order.

Solution

</> Code

MySQL  Auto

```
1 SELECT user_id, name, mail
2 FROM Users
3 WHERE REGEXP_LIKE(
4     mail,
5     '^[A-Za-z][A-Za-z0-9._-]*@leetcode\\.com$',
6     'c'
7 );
```

Accepted Runtime: 98 ms

 Case 1

Input

Users =

user_id	name	mail
1	Winston	winston@leetcode.com
2	Jonathan	jonathanisgreat
3	Annabelle	bella@leetcode.com
4	Sally	sally.come@leetcode.com
5	Marwan	quarz#2020@leetcode.com
6	David	david69@gmail.com
7	Shapiro	.shapo@leetcode.com

Output

user_id	name	mail
1	Winston	winston@leetcode.com
3	Annabelle	bella@leetcode.com
4	Sally	sally.come@leetcode.com

Explanation

- REGEXP – Used for pattern matching in MySQL.
- ^ – Ensures the email starts with a letter
- * – Represents zero or more repetitions.
- @leetcode.com – Ensures the email ends with @leetcode.com.
- \$ – Ensures the email ends exactly at .com.

2. Patients With a Condition

Problem Summary

You are given a table named Patients.

Column Name	Type
patient_id	int
patient_name	Varchar
conditions	Varchar

- patient_id is the primary key.

Important Rule

A patient is considered valid if:

- DIAB1 appears at the beginning of the string, or
- DIAB1 appears after a space.

Task

Write an SQL query to report the patient ID, patient name, and conditions of patients who have Type I Diabetes (DIAB1)

Return the result table in any order.

Solution

</> Code

MySQL   Auto

```
1 SELECT
2     patient_id,
3     patient_name,
4     conditions
5 FROM Patients
6 WHERE conditions LIKE 'DIAB1%'
7     OR conditions LIKE '% DIAB1%'
```

Accepted Runtime: 77 ms

Case 1

Input

Patients =

patient_id	patient_name	conditions
1	Daniel	YFEV COUGH
2	Alice	
3	Bob	DIAB100 MYOP
4	George	ACNE DIAB100
5	Alain	DIAB201

Output

patient_id	patient_name	conditions
3	Bob	DIAB100 MYOP
4	George	ACNE DIAB100

Explanation

- Bob and George have DIAB1 in their conditions.
- DIAB1 represents Type I Diabetes.
- The query checks whether DIAB1 appears at the beginning or after a space.
- Therefore, those patients are displayed in the result.

3.Second Highest Salary

Problem Summary

You are given an Employee table.

Column Name	Type
id	int
salary	int

- id is the primary key.
- Each row contains the salary of an employee.
- Salaries may contain duplicates.

Your task is to find the second highest distinct salary.

If there is no second highest salary, return NULL.

Task

Write an SQL query to report the second highest distinct salary from the Employee table.

Solution

</> Code

MySQL  Auto

```
1 SELECT
2 (
3     SELECT DISTINCT salary
4     FROM Employee
5     ORDER BY salary DESC
6     LIMIT 1 OFFSET 1
7 ) AS SecondHighestSalary;
```

Accepted Runtime: 87 ms

☒ Case 1 ☒ Case 2

Input

Employee =

id	salary
1	100
2	200
3	300

Output

SecondHighestSalary
200

Explanation

Distinct salaries:

100, 200, 300

Highest salary = 300

Second highest salary = 200

4.Customer Placing the Largest Number of Orders

Problem Summary

You are given an Orders table.

Column Name	Type
order_ number	int
customer_ number	int

- order_ number is unique.
- Each row represents one order placed by a customer.
- A customer can place multiple orders.

Your task is to find the customer **who placed the largest number of orders**.

The problem guarantees that:

There is exactly **one customer** who has placed the maximum number of orders.

Task

Write an SQL query to find the customer_ number of the customer who has placed the largest number of orders.

Return the result table in any order.

Solution

 Code

MySQL   Auto

```
1 SELECT customer_number
2 FROM Orders
3 GROUP BY customer_number
4 ORDER BY COUNT(*) DESC
5 LIMIT 1;
```

Accepted Runtime: 64 ms

Case 1

Input

```
orders =  
| order_number | customer_number |  
|-----|-----|  
| 1           | 1           |  
| 2           | 2           |  
| 3           | 3           |  
| 4           | 3           |
```

Output

```
| customer_number |  
|-----|  
| 3           |
```

Explanation

Count the orders placed by each customer:

customer_number	Number of Orders
1	1
2	1
3	2

5.Product Sales Analysis III

Problem Summary

You are given two tables, but only the Sales table is needed for this problem.

Sales Table

Column Name	Type
sale_id	int
product_id	int
year	int
quantity	int
price	int

- (sale_id, year) is the primary key.
- Each row represents a sale of a product in a particular Year.

Objective

For **each product**, find:

1. The **first year** in which the product was sold.
2. The **quantity** sold in that year.
3. The **price** of the product in that year.

Task

Write an SQL query to repost:

- Product_id
- First_year
- Quantity
- Price

For the **first year** in which each product was sold.

Return the result table in any order.

Solution

```
</>Code
MySQL  Auto
1  SELECT
2      product_id,
3      year first_year,
4      quantity,
5      price
6  FROM Sales
7  WHERE (product_id, year) IN (SELECT product_id, MIN(year)
8                                FROM Sales
9                                GROUP BY product_id)
```

Accepted Runtime: 71 ms

Case 1

Input

Sales =

sale_id	product_id	year	quantity	price
1	100	2008	10	5000
2	100	2009	12	5000
7	200	2011	15	9000

Output

product_id	first_year	quantity	price
100	2008	10	5000
200	2011	15	9000

Explanation

The query finds the first year each product was sold.

- It checks each product_id in the Sales table.
- MIN(year) finds the first year the product was sold.
- Then it gets the quantity and price for that first year.
- The result shows product_id, first_year, quantity, and price.
- So, each product appears with its earliest sale details.